

Orientation-Augmented Single-Molecule Tracking via 6D-SMOLM Cost Function Integration within u-track3D

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Abstract

Single-molecule orientation tracking requires joint optimization of spatial and angular coordinates, a capability absent from existing particle tracking frameworks. We present a wrapping strategy that extends u-track3D [1] with orientation-aware cost functions for the linear assignment problem (LAP), enabling six-dimensional single-molecule orientation localization microscopy (6D-SMOLM) tracking. The implementation preserves the validated spatial Kalman filter [2] of u-track3D while augmenting the linking and gap-closing cost matrices with angular distance terms derived from rotational Brownian dynamics on the unit sphere. Three orientation motion models compete at each linking step, and per-track rotational diffusion coefficients are learned adaptively from track history. This approach resolves linking ambiguities that purely spatial tracking cannot, particularly in dense fields where molecules with distinct orientations occupy similar spatial positions (Figure 2).

1. Introduction and Software Architecture

The 6D-SMOLM tracking pipeline extends u-track3D through a wrapper architecture that preserves the original spatial tracking engine while injecting orientation information into the cost matrices. In u-track3D, two sequential linear assignment problems (LAPs) govern track construction [1]: LAP 1 performs frame-to-frame linking by optimizing assignments between detections in consecutive frames, and LAP 2 closes gaps, merges, and splits track segments across multiple frames (Figure 2). Our wrapper functions, `costMat6DSMOLMLink` and `costMat6DSMOLMCloseGaps`, first call the original u-track3D cost functions (`costMatRandomDirectedSwitchingMotionLink` and `costMatRandomDirectedSwitchingMotionCloseGaps`) to obtain baseline spatial cost matrices, then augment each valid cost entry with an additive orientation penalty. This strategy ensures that the spatial Kalman filter state propagation, nearest-neighbor distance computation, and all other internal u-track3D bookkeeping remain unmodified, isolating orientation logic to the cost function layer alone.

Each 6D-SMOLM localization provides six coordinates: three spatial (x, y, z) and three orientational (θ, ϕ, ω). The polar angle θ and azimuthal angle ϕ define the mean dipole orientation unit vector on the upper hemisphere, while the wobble angle ω quantifies the solid angle subtended by orientational fluctuations during the camera exposure [3]. The spatial coordinates enter the cost matrix through u-track3D's existing Kalman filter machinery, while θ and ϕ enter through our orientation cost term. The wobble angle ω serves a dual role: it contributes an optional independent cost term and modulates the measurement noise assigned to θ and ϕ , since higher wobble degrades the precision of instantaneous orientation estimates [4].

2. Spatial Tracking via Kalman Filtering

The u-track3D Kalman filter provides optimal position predictions by recursively estimating particle state vectors from noisy localizations [2]. For each tracked particle, the state vector contains position and velocity components in each spatial dimension (x, y, z). At each frame transition, the filter performs a

prediction step: the state is propagated forward using a linear motion model, and the state covariance is expanded to account for process noise arising from the stochastic component of particle motion. The prediction equations are:

$$\begin{aligned}\hat{x}(t|t-1) &= F \cdot \hat{x}(t-1|t-1) \\ P(t|t-1) &= F \cdot P(t-1|t-1) \cdot F^T + Q\end{aligned}$$

Here F is the state transition matrix encoding the assumed motion model (constant velocity with random acceleration), P is the state error covariance matrix, and Q is the process noise covariance [2]. u-track3D supports multiple competing motion models (e.g., directed, Brownian, confined) and selects the model yielding the lowest cost for each track-detection pair [1].

The spatial linking cost between a predicted track position and a candidate detection is the squared Mahalanobis distance [5], which normalizes the position residual by the total uncertainty. The innovation vector y is the difference between the observed detection position z and the predicted position [2], and the innovation covariance S combines the prediction uncertainty with the measurement noise R :

$$\begin{aligned}y &= z - H \cdot \hat{x}(t|t-1) \\ S &= H \cdot P(t|t-1) \cdot H^T + R \\ d_{spatial}^2 &= y^T \cdot S^{-1} \cdot y\end{aligned}$$

The observation matrix H extracts the position components from the full state vector. This Mahalanobis distance naturally accounts for anisotropic uncertainties [5]: a particle with high velocity uncertainty in one direction will have a proportionally larger search radius along that axis. The spatial cost matrix entry $C_{(spatial)}(i,j)$ between track i and detection j is this squared Mahalanobis distance, with entries set to a `nonlinkMarker` value (effectively infinity) for pairs exceeding a maximum search radius [1].

3. Orientation Cost Function: Adaptive Three-Model Kalman (Level 3)

3.1 Angular Distance with Dipole Symmetry

The angular distance between two dipole orientations must respect the head-tail symmetry inherent to fluorescent dipole emitters [3,4]. A fluorescent dipole oriented at angles (θ, φ) is physically indistinguishable from one at $(\pi-\theta, \varphi+\pi)$, producing identical emission patterns in single-molecule imaging. To enforce this symmetry, we convert the spherical angles to unit vectors on the upper hemisphere and define the angular distance through the absolute value of their dot product:

$$\hat{\mu} = (\sin\theta \cos\varphi, \quad \sin\theta \sin\varphi, \quad \cos\theta)(|\hat{\mu}_1 \cdot \hat{\mu}_2|)$$

The resulting angular distance ψ lies in the range $[0, \pi/2]$, correctly identifying antipodal orientations as identical. This is the fundamental distance metric used throughout the orientation cost computation.

3.2 Rotational Brownian Dynamics on S^2

The expected angular displacement of a freely rotating dipole follows from rotational Brownian dynamics on the unit sphere S^2 [6]. On S^2 , the orientation diffuses along two independent tangent-plane directions, each contributing a variance of $2D_{(rot)}\Delta t$ per time step, where $D_{(rot)}$ is the rotational diffusion coefficient. The total mean-squared angular displacement therefore involves two components:

$$E[\Delta\psi^2] = 4D_{rot}\Delta t$$

This factor of 4 (rather than 2) distinguishes rotational diffusion on S^2 from one-dimensional translational diffusion [6]. It is the rotational analog of the well-known result that mean-squared displacement in d dimensions equals $2dDt$.

3.3 Three Competing Motion Models

Level 3 orientation tracking evaluates three competing motion models for each track-detection candidate pair and selects the model yielding the lowest cost. Given a track's current orientation and its estimated angular velocity from the previous two frames, the three models predict distinct orientations for the next frame. The forward model propagates the orientation by rotating the current direction along the estimated angular velocity axis. The backward model rotates by the same magnitude in the opposite direction. The Brownian model predicts no net rotation, expecting the orientation to remain unchanged. This multi-model approach parallels u-track3D's own strategy of competing spatial motion models [1].

Orientation propagation uses the Rodrigues rotation formula with the sign convention of Hofling and Straube [6]:

$$\hat{\mu}_{rot} = \hat{\mu}\cos\alpha - (\hat{k} \times \hat{\mu})\sin\alpha + \hat{k}(\hat{k} \cdot \hat{\mu})(1 - \cos\alpha)$$

is the unit rotation axis and α is the rotation angle.

The negative sign on the cross-product term follows the convention used in the synthetic trajectory generator (generate6DTrajectories.m), ensuring consistency between simulation and tracking. The angular velocity is estimated from the last two frames of track history by computing the Rodrigues axis and angle that rotate the previous orientation into the current one.

3.4 Per-Track $D_{(rot)}$ Learning

The rotational diffusion coefficient $D_{(rot)}$ is learned adaptively for each track from its angular displacement history, following an approach analogous to maximum-likelihood estimation of translational diffusion coefficients [7]. For a track with at least three frames of history, the angular displacements between consecutive detections are computed, and $D_{(rot)}$ is estimated from the mean-squared displacement with a correction for measurement noise:

$$D_{rot} = \frac{\langle \Delta\alpha^2 \rangle - 2\sigma_\psi^2}{4\Delta t}$$

The term $2\sigma_\psi^2$ subtracts the contribution of measurement localization error, where σ_ψ is the orientation Cramer-Rao lower bound (CRLB) [4]. The factor of 4 in the denominator accounts for the two tangent-plane components of rotational diffusion on S^2 [6]. The estimated $D_{(rot)}$ is clamped to the range $[0.001, 10 \times D_{(rot,prior)}]$ to prevent divergence from noisy short tracks, where $D_{(rot,prior)}$ is the user-supplied initial estimate. Tracks shorter than the minimum history length (default: 3 frames) use $D_{(rot,prior)}$.

3.5 Orientation Innovation Covariance and Cost

The orientation cost mirrors the spatial Mahalanobis distance [5] by normalizing the squared angular displacement by an innovation covariance [2]. The orientation innovation covariance S_ψ combines the process variance (predicted angular spread) with the measurement variances of both the current track endpoint and the candidate detection:

$$P_{process} = 4D_{rot}\Delta t$$

$$S_\psi = P_{process} + R_{curr} + R_{next}$$

Here R_{curr} and R_{next} are the squared measurement uncertainties for the current and candidate orientations, respectively. When omega-dependent weighting is enabled, these measurement variances are modulated by the wobble angle through the order parameter γ [3]: larger wobble (lower γ) increases R , reflecting the reduced angular precision of a rapidly wobbling dipole [4]. The orientation cost under each motion model m is:

$$C_{orient,m} = \frac{\Delta\psi_m^2}{S_\psi}$$

Level 3 selects the minimum cost across all three models:

$$C_{orient} = \min(C_{orient,fwd}, C_{orient,bwd}, C_{orient,brn})$$

The winning model index is recorded in the `orientPropagationScheme` matrix for diagnostic purposes, allowing post hoc analysis of which motion regime dominates in different regions of the sample.

4. Combined Cost Matrix and LAP Assignment

The total linking cost combines spatial and orientation contributions through user-specified weights, enabling the LAP solver to simultaneously optimize positional and orientational consistency [1]. For each valid track-detection pair (i, j) in the linking cost matrix:

$$C_{(total)(i,j)} = w_{spatial} \cdot d_{spatial}^{2(i,j)} + w_{orient} \cdot C_{(orient)(i,j)} + w_\omega \cdot C_\omega(i,j)$$

The weight $w_{spatial}$ (default 1.0) scales the spatial Mahalanobis distance [5], w_{orient} (default 1.0) scales the orientation cost, and w_ω (default 0.0) scales an optional wobble continuity cost. The wobble cost C_ω penalizes changes in the wobble solid angle ω between linked detections. Only the upper-left block of the cost matrix (valid linking region) receives the orientation augmentation; the birth, death, and auxiliary blocks remain unmodified from the base `u-track3D` output [1].

The augmented cost matrix is solved by the Jonker-Volgenant algorithm [8] to obtain the globally optimal assignment. This algorithm solves the linear assignment problem in $O(n^3)$ time, simultaneously minimizing the total spatial and orientational linking cost across all particles, subject to the constraint that each detection is assigned to at most one track [1]. Unmatched tracks are terminated (assigned to death dummies) and unmatched detections initiate new tracks (birth dummies). The key advantage of the additive cost formulation is that orientation information naturally resolves spatial ambiguities: when two tracks approach the same detection, the orientation cost breaks the tie in favor of the track with compatible angular coordinates (Figure 2A).

4.1 Gap Closing with Orientation

The second LAP closes gaps between track segments using orientation similarity as additional evidence for segment identity [1]. When a track segment ends at frame t_{end} and another begins at frame t_{start} (with $t_{start} > t_{end} + 1$), the gap-closing cost combines spatial proximity with angular distance between the endpoint and startpoint orientations. The angular distance is normalized by the square root of the time gap to account for expected rotational diffusion during the gap interval, since the mean angular

displacement scales as $\sqrt{(4D_{\text{rot}})\Delta t}$ under Brownian rotation [6]. Three scenarios illustrate the value of orientation in gap closing (Figure 2B): (i) a track gap is correctly closed when the endpoint and startpoint share compatible orientations; (ii) a gap closure is rejected when orientations mismatch despite spatial proximity; and (iii) crossing tracks that are spatially ambiguous are correctly resolved by their distinct orientation signatures.

Figure 2

[INSERT FIGURE 2 PANELS A AND B HERE]

Figure 2. Orientation-augmented linear assignment problems for 6D-SMOLM tracking. (A) Frame-to-frame linking (LAP 1). Three particles (A, B, C) in frame t are linked to detections in frame $t+1$. Colored arrows at each detection represent the measured dipole orientation (θ, ϕ) . Solid colored lines indicate optimal 6D assignments; dashed black lines indicate rejected candidate links. Orientation similarity between matched pairs (e.g., A to A') reduces the total cost below that of spatially similar but orientationally incompatible alternatives. (B) Gap closing and segment linking (LAP 2) via orientational coordinates. Three scenarios are shown: (i) correct gap closing when the segment endpoint and startpoint share a low-cost orientation match; (ii) rejection of a segment link due to (θ, ϕ) mismatch despite spatial continuity; and (iii) disambiguation of crossing trajectories where high spatial costs alone are insufficient, but distinct orientation signatures correctly assign detections to the appropriate track segments.

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